QuantiumTask2

March 2, 2026

1 Importing The necessary libraries

[1]:
```
import pandas as pd;
import matplotlib.pyplot as plt;
import numpy as np ;
```

2 Importing the QVI_Data file

[2]:
```
qvi_data = pd.read_csv("C:/Users/user/OneDrive - Hetauda School of Management␣
↪and Social Sciences/Desktop/Data Analysis/Projects/Project III/Raw Data &␣
↪Problems/QVI_data_web.csv");
qvi_data.head()
```

[2]:
```
   LYLTY_CARD_NBR        DATE  STORE_NBR  TXN_ID  PROD_NBR  \
0            1000  10/17/2018          1       1         5
1            1002   9/16/2018          1       2        58
2            1003    3/7/2019          1       3        52
3            1003    3/8/2019          1       4       106
4            1004   11/2/2018          1       5        96

                                PROD_NAME  PROD_QTY  TOT_SALES  PACK_SIZE  \
0  Natural Chip        Compny SeaSalt175g         2        6.0        175
1   Red Rock Deli Chikn&Garlic Aioli 150g         1        2.7        150
2   Grain Waves Sour    Cream&Chives 210G         1        3.6        210
3  Natural ChipCo      Hony Soy Chckn175g         1        3.0        175
4         WW Original Stacked Chips 160g         1        1.9        160

        BRAND              LIFESTAGE PREMIUM_CUSTOMER
0     NATURAL  YOUNG SINGLES/COUPLES          Premium
1         RRD  YOUNG SINGLES/COUPLES       Mainstream
2     GRNWVES         YOUNG FAMILIES           Budget
3     NATURAL         YOUNG FAMILIES           Budget
4  WOOLWORTHS  OLDER SINGLES/COUPLES       Mainstream
```

3 Adding new column as YEARMONTH

[3]:
```
qvi_data["DATE"] = pd.to_datetime(qvi_data["DATE"]);
qvi_data["YEARMONTH"] = qvi_data["DATE"].dt.year * 100 + qvi_data["DATE"].dt.
 ↪month ;
qvi_data.head()
```

[3]:
```
   LYLTY_CARD_NBR       DATE  STORE_NBR  TXN_ID  PROD_NBR  \
0            1000 2018-10-17          1       1         5
1            1002 2018-09-16          1       2        58
2            1003 2019-03-07          1       3        52
3            1003 2019-03-08          1       4       106
4            1004 2018-11-02          1       5        96

                                PROD_NAME  PROD_QTY  TOT_SALES  PACK_SIZE  \
0  Natural Chip        Compny SeaSalt175g         2        6.0        175
1   Red Rock Deli Chikn&Garlic Aioli 150g         1        2.7        150
2   Grain Waves Sour    Cream&Chives 210G         1        3.6        210
3  Natural ChipCo      Hony Soy Chckn175g         1        3.0        175
4          WW Original Stacked Chips 160g         1        1.9        160

        BRAND              LIFESTAGE PREMIUM_CUSTOMER  YEARMONTH
0     NATURAL  YOUNG SINGLES/COUPLES          Premium     201810
1         RRD  YOUNG SINGLES/COUPLES       Mainstream     201809
2     GRNWVES         YOUNG FAMILIES           Budget     201903
3     NATURAL         YOUNG FAMILIES           Budget     201903
4  WOOLWORTHS  OLDER SINGLES/COUPLES       Mainstream     201811
```

4 Defining Metrics

[4]:
```
measureOverTime = (qvi_data.groupby(["STORE_NBR", "YEARMONTH"]).agg(
    totSales=("TOT_SALES","sum"),
    ntransactions=("TXN_ID","nunique"),
    ncustomers=("LYLTY_CARD_NBR","nunique"),
    totalQty=("PROD_QTY","sum")
).reset_index()
);

measureOverTime["nTxnPerCust"] = (measureOverTime["ntransactions"] /␣
 ↪measureOverTime["ncustomers"]);

measureOverTime["nChipsPerTxn"] = (measureOverTime["totalQty"]/
 ↪measureOverTime["ntransactions"]);

measureOverTime["avgPricePerUnit"] = (measureOverTime["totSales"]/
 ↪measureOverTime["totalQty"]) ;
```

```
measureOverTime
```

[4]:

```
      STORE_NBR  YEARMONTH  totSales  ntransactions  ncustomers  totalQty  \
0             1     201807     206.9             52          49        62
1             1     201808     176.1             43          42        54
2             1     201809     278.8             62          59        75
3             1     201810     188.1             45          44        58
4             1     201811     192.6             47          46        57
…           …         …        …              …           …        …
3164        272     201902     395.5             48          45        91
3165        272     201903     442.3             53          50       101
3166        272     201904     445.1             55          54       105
3167        272     201905     314.6             40          34        71
3168        272     201906     312.1             37          34        70

      nTxnPerCust  nChipsPerTxn  avgPricePerUnit
0        1.061224      1.192308         3.337097
1        1.023810      1.255814         3.261111
2        1.050847      1.209677         3.717333
3        1.022727      1.288889         3.243103
4        1.021739      1.212766         3.378947
…           …            …               …
3164     1.066667      1.895833         4.346154
3165     1.060000      1.905660         4.379208
3166     1.018519      1.909091         4.239048
3167     1.176471      1.775000         4.430986
3168     1.088235      1.891892         4.458571

[3169 rows x 9 columns]
```

5 Filtering Pre Trial Months

[5]:

```
n_pretrial_months = measureOverTime[
    measureOverTime["YEARMONTH"] < 201902
]["YEARMONTH"].nunique()

print("Number of pre-trial months:", n_pretrial_months)

preTrialCounts = (
    measureOverTime[measureOverTime["YEARMONTH"] < 201902]
    .groupby("STORE_NBR")["YEARMONTH"]
    .nunique()
    .reset_index(name="month_count")
) ;

storesWithFullObs = preTrialCounts[
```

```
    preTrialCounts["month_count"] == n_pretrial_months
]["STORE_NBR"]

preTrialMeasures = measureOverTime[
    (measureOverTime["YEARMONTH"] < 201902) &
    (measureOverTime["STORE_NBR"].isin(storesWithFullObs))
]

preTrialMeasures
```

```
Number of pre-trial months: 7
```

[5]:

```
      STORE_NBR  YEARMONTH  totSales  ntransactions  ncustomers  totalQty  \
0             1     201807     206.9             52          49        62
1             1     201808     176.1             43          42        54
2             1     201809     278.8             62          59        75
3             1     201810     188.1             45          44        58
4             1     201811     192.6             47          46        57
…           …         …        …             …           …         …
3159        272     201809     304.7             36          32        71
3160        272     201810     430.6             50          44        99
3161        272     201811     376.2             45          41        87
3162        272     201812     403.9             47          47        89
3163        272     201901     423.0             50          46        96

      nTxnPerCust  nChipsPerTxn  avgPricePerUnit
0        1.061224      1.192308         3.337097
1        1.023810      1.255814         3.261111
2        1.050847      1.209677         3.717333
3        1.022727      1.288889         3.243103
4        1.021739      1.212766         3.378947
…           …            …               …
3159     1.125000      1.972222         4.291549
3160     1.136364      1.980000         4.349495
3161     1.097561      1.933333         4.324138
3162     1.000000      1.893617         4.538202
3163     1.086957      1.920000         4.406250

[1820 rows x 9 columns]
```

6 Calculation of Correlation of store 77,86 & 88

[6]:

```
def calculate_correlation(inputTable, metricCol, storeComparison):

    trial_store = inputTable[inputTable["STORE_NBR"] == storeComparison]

    storeNumbers = inputTable["STORE_NBR"].unique()
```

```
    storeNumbers = storeNumbers[storeNumbers != storeComparison]

    results = []

    for store in storeNumbers:

        control_store = inputTable[inputTable["STORE_NBR"] == store]

        merged = trial_store.merge(
            control_store,
            on="YEARMONTH",
            suffixes=("_trial", "_control")
        )

        # Calculate correlation
        corr = merged[f"{metricCol}_trial"].corr(
            merged[f"{metricCol}_control"]
        )

        results.append({
            "Store1": storeComparison,
            "Store2": store,
            "corr_measure": corr
        })

    return pd.DataFrame(results)

corr_sales_77 = calculate_correlation(
    preTrialMeasures,
    "totSales",
    77
)
corr_sales_84 = calculate_correlation(
    preTrialMeasures,
    "totSales",
    84
)
corr_sales_88 = calculate_correlation(
    preTrialMeasures,
    "totSales",
    88
)
print(corr_sales_77.sort_values("corr_measure", ascending=False).head())
print()
print(corr_sales_84.sort_values("corr_measure", ascending=False).head())
print()
print(corr_sales_88.sort_values("corr_measure", ascending=False))
```

```
     Store1  Store2  corr_measure
67       77      71      0.914106
220      77     233      0.903774
110      77     119      0.867664
15       77      17      0.842668
2        77       3      0.806644

     Store1  Store2  corr_measure
159      84     168      0.953761
148      84     157      0.833558
187      84     197      0.818319
2        84       3      0.804775
189      84     199      0.799090

     Store1  Store2  corr_measure
150      88     159      0.903186
194      88     204      0.885774
125      88     134      0.864293
0        88       1      0.813636
239      88     253      0.811838
..      …      …           …
258      88     272     -0.772772
21       88      23     -0.801652
7        88       8     -0.816296
44       88      48     -0.857142
217      88     230     -0.908883

[259 rows x 3 columns]
```

7 For trial store 77

8 Calculating Correlation and Magnitudes

[7]:
```
def calculate_magnitude_distance(inputTable, metricCol, storeComparison):

    trial_store = inputTable[inputTable["STORE_NBR"] == storeComparison]

    storeNumbers = inputTable["STORE_NBR"].unique()
    storeNumbers = storeNumbers[storeNumbers != storeComparison]

    results = []

    for store in storeNumbers:

        control_store = inputTable[inputTable["STORE_NBR"] == store]
```

```
        merged = trial_store.merge(
            control_store,
            on="YEARMONTH",
            suffixes=("_trial", "_control")
        )

        # Absolute difference
        merged["measure"] = abs(
            merged[f"{metricCol}_trial"] -
            merged[f"{metricCol}_control"]
        )

        # Min-Max scaling per month
        min_dist = merged["measure"].min()
        max_dist = merged["measure"].max()

        if max_dist - min_dist == 0:
            merged["magnitudeMeasure"] = 1
        else:
            merged["magnitudeMeasure"] = 1 - (
                (merged["measure"] - min_dist) /
                (max_dist - min_dist)
            )

        final_score = merged["magnitudeMeasure"].mean()

        results.append({
            "Store1": storeComparison,
            "Store2": store,
            "mag_measure": final_score
        })

    return pd.DataFrame(results)

trial_store = 77

# Correlation
corr_nSales = calculate_correlation(preTrialMeasures, "totSales", trial_store)
corr_nCustomers = calculate_correlation(preTrialMeasures, "ncustomers",␣
 ↪trial_store)

# Magnitude
magnitude_nSales = calculate_magnitude_distance(preTrialMeasures, "totSales",␣
 ↪trial_store)
magnitude_nCustomers = calculate_magnitude_distance(preTrialMeasures,␣
 ↪"ncustomers", trial_store)
```

```python
combined = (
    corr_nSales.rename(columns={"corr_measure": "corr_sales"})
    .merge(
        corr_nCustomers.rename(columns={"corr_measure": "corr_customers"}),
        on=["Store1", "Store2"]
    )
    .merge(
        magnitude_nSales,
        on=["Store1", "Store2"]
    )
    .merge(
        magnitude_nCustomers,
        on=["Store1", "Store2"],
        suffixes=("_sales", "_customers")
    )
)

combined["final_score"] = (
    combined["corr_sales"] +
    combined["corr_customers"] +
    combined["mag_measure_sales"] +
    combined["mag_measure_customers"]
) / 4

combined.sort_values("final_score", ascending=False)
```

[7]:

```
     Store1  Store2  corr_sales  corr_customers  mag_measure_sales  \
220      77     233    0.903774        0.990358           0.490806
67       77      71    0.914106        0.754817           0.664887
78       77      84    0.684348        0.858571           0.629596
110      77     119    0.867664        0.983267           0.404428
107      77     115    0.689159        0.718882           0.727536
..      …      …          …               …                  …
160      77     169   -0.630112       -0.666391           0.473506
229      77     242   -0.692664       -0.643351           0.449028
177      77     186   -0.820214       -0.635966           0.562601
94       77     102   -0.550834       -0.652527           0.346357
8        77       9   -0.702976       -0.785699           0.414734

     mag_measure_customers  final_score
220               0.642857     0.756949
67                0.571429     0.726310
78                0.571429     0.685986
110               0.476190     0.682887
107               0.571429     0.676751
..                     …          …
160               0.523810    -0.074797
```

```
229                0.550000    -0.084247
177                0.521429    -0.093037
94                 0.414286    -0.110680
8                  0.611429    -0.115628

[259 rows x 7 columns]
```

9 Combining magnitude and correlation

[8]:
```
corr_weight = 0.5

score_nSales = (
    corr_nSales
    .merge(magnitude_nSales, on=["Store1", "Store2"])
)

score_nSales["scoreNSales"] = (
    corr_weight * score_nSales["corr_measure"] +
    (1 - corr_weight) * score_nSales["mag_measure"]
)

score_nCustomers = (
    corr_nCustomers
    .merge(magnitude_nCustomers, on=["Store1", "Store2"])
)

score_nCustomers["scoreNCust"] = (
    corr_weight * score_nCustomers["corr_measure"] +
    (1 - corr_weight) * score_nCustomers["mag_measure"]
)

score_Control = (
    score_nSales[["Store1", "Store2", "scoreNSales"]]
    .merge(
        score_nCustomers[["Store1", "Store2", "scoreNCust"]],
        on=["Store1", "Store2"]
    )
)

score_Control["finalControlScore"] = (
    0.5 * score_Control["scoreNSales"] +
    0.5 * score_Control["scoreNCust"]
)

score_Control_sorted = score_Control.sort_values(
    "finalControlScore",
```

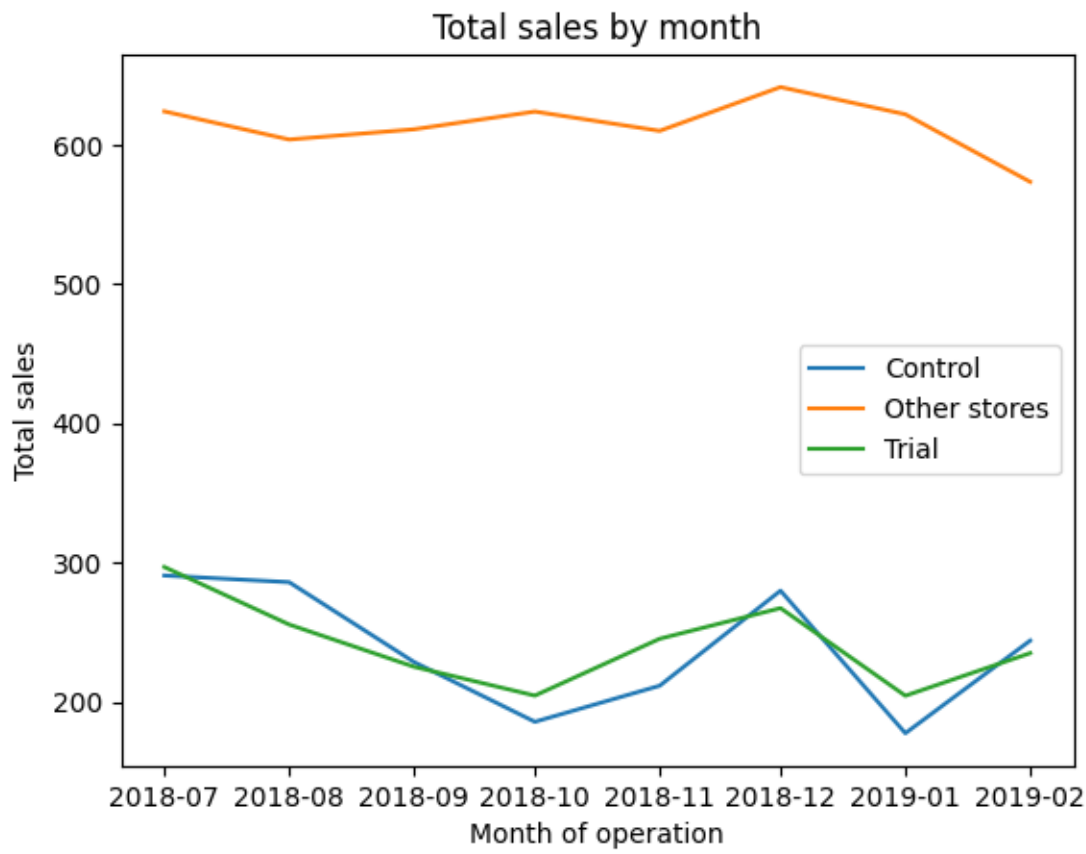
```
    ascending=False
)

score_Control_sorted.head()

control_store = score_Control_sorted.iloc[0]["Store2"]
control_store
```

[8]: np.float64(233.0)

10 Visual check of TotalSales

[9]:
```
trial_store = 77
control_store = control_store

measureOverTimeSales = measureOverTime.copy()

measureOverTimeSales["Store_type"] = np.where(
    measureOverTimeSales["STORE_NBR"] == trial_store,
    "Trial",
    np.where(
        measureOverTimeSales["STORE_NBR"] == control_store,
        "Control",
        "Other stores"
    )
)

pastSales = (
    measureOverTimeSales
    .groupby(["YEARMONTH", "Store_type"])["totSales"]
    .mean()
    .reset_index()
)

pastSales["TransactionMonth"] = pd.to_datetime(
    pastSales["YEARMONTH"].astype(str) + "01",
    format="%Y%m%d"
)

pastSales = pastSales[pastSales["YEARMONTH"] < 201903]

plt.figure()
for label in pastSales["Store_type"].unique():
    subset = pastSales[pastSales["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["totSales"],  label=label)

plt.xlabel("Month of operation")
```

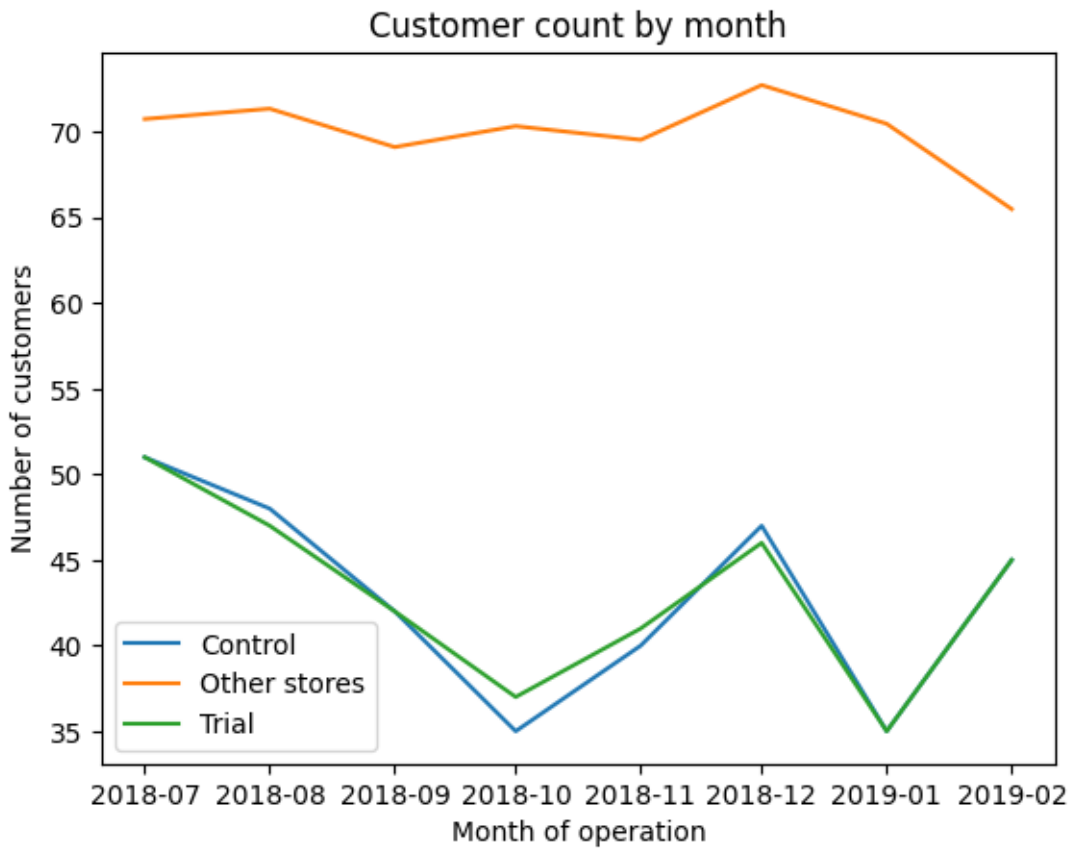
```
plt.ylabel("Total sales")
plt.legend()
plt.title("Total sales by month")
plt.show()
```



11 Visual check of No of Customers

[10]:
```
measureOverTimeCusts = measureOverTime.copy()

measureOverTimeCusts["Store_type"] = np.where(
    measureOverTimeCusts["STORE_NBR"] == trial_store,
    "Trial",
    np.where(
        measureOverTimeCusts["STORE_NBR"] == control_store,
        "Control",
        "Other stores"
    )
)
```

```
measureOverTimeCusts

pastCustomers = (
    measureOverTimeCusts
    .groupby(["YEARMONTH", "Store_type"])["ncustomers"]
    .mean()
    .reset_index()
)

pastCustomers["TransactionMonth"] = pd.to_datetime(
    pastCustomers["YEARMONTH"].astype(str) + "01",
    format="%Y%m%d"
)

pastCustomers = pastCustomers[pastCustomers["YEARMONTH"] < 201903]

plt.figure()
for label in pastCustomers["Store_type"].unique():
    subset = pastCustomers[pastCustomers["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["ncustomers"], label=label )

plt.xlabel("Month of operation")
plt.ylabel("Number of customers")
plt.legend()
plt.title("Customer count by month")
plt.show()
```



12 Checking if sales significantly increase during Feb–Apr 2019

13 Computing scaling factors { Sales }

[11]:
```
trial_pre_sales = preTrialMeasures[
    preTrialMeasures["STORE_NBR"] == trial_store
]["totSales"].sum()

control_pre_sales = preTrialMeasures[
    preTrialMeasures["STORE_NBR"] == control_store
]["totSales"].sum()

scalingFactorForControlSales = trial_pre_sales / control_pre_sales

scalingFactorForControlSales
```

[11]: np.float64(1.023617303289553)

14 Applying Scaling to Control Store

[12]:
```
measureOverTimeSales = measureOverTime.copy()

scaledControlSales = measureOverTimeSales[
    measureOverTimeSales["STORE_NBR"] == control_store
].copy()

scaledControlSales["controlSales"] = (
    scaledControlSales["totSales"] * scalingFactorForControlSales
)
scaledControlSales
```

[12]:

	STORE_NBR	YEARMONTH	totSales	ntransactions	ncustomers	totalQty	\
2699	233	201807	290.7	54	51	88	
2700	233	201808	285.9	50	48	80	
2701	233	201809	228.6	45	42	70	
2702	233	201810	185.7	36	35	56	
2703	233	201811	211.6	41	40	62	
2704	233	201812	279.8	50	47	75	
2705	233	201901	177.5	35	35	47	
2706	233	201902	244.0	47	45	70	
2707	233	201903	199.1	41	40	59	
2708	233	201904	158.6	32	30	46	
2709	233	201905	344.4	62	57	92	
2710	233	201906	221.0	41	41	61	

	nTxnPerCust	nChipsPerTxn	avgPricePerUnit	controlSales
2699	1.058824	1.629630	3.303409	297.565550
2700	1.041667	1.600000	3.573750	292.652187
2701	1.071429	1.555556	3.265714	233.998916
2702	1.028571	1.555556	3.316071	190.085733
2703	1.025000	1.512195	3.412903	216.597421
2704	1.063830	1.500000	3.730667	286.408121
2705	1.000000	1.342857	3.776596	181.692071
2706	1.044444	1.489362	3.485714	249.762622
2707	1.025000	1.439024	3.374576	203.802205
2708	1.066667	1.437500	3.447826	162.345704
2709	1.087719	1.483871	3.743478	352.533799
2710	1.000000	1.487805	3.622951	226.219424

15 Merging Trial + Scaled Control

[13]:
```
trialSales = measureOverTimeSales[
    measureOverTimeSales["STORE_NBR"] == trial_store
][["YEARMONTH", "totSales"]]
```

```
percentageDiff = trialSales.merge(
    scaledControlSales[["YEARMONTH", "controlSales"]],
    on="YEARMONTH"
)
percentageDiff
```

[13]:

	YEARMONTH	totSales	controlSales
0	201807	296.8	297.565550
1	201808	255.5	292.652187
2	201809	225.2	233.998916
3	201810	204.5	190.085733
4	201811	245.3	216.597421
5	201812	267.3	286.408121
6	201901	204.4	181.692071
7	201902	235.0	249.762622
8	201903	278.5	203.802205
9	201904	263.5	162.345704
10	201905	299.3	352.533799
11	201906	264.7	226.219424

16 Calculating Percentage Difference & Standard Deviation on pretrial only

[14]:

```
percentageDiff["percentageDiff"] = (
    (percentageDiff["totSales"] - percentageDiff["controlSales"])
    / percentageDiff["controlSales"]
);

stdDev = percentageDiff[
    percentageDiff["YEARMONTH"] < 201902
]["percentageDiff"].std();

print(stdDev);
percentageDiff
```

```
0.09958646884078388
```

[14]:

	YEARMONTH	totSales	controlSales	percentageDiff
0	201807	296.8	297.565550	-0.002573
1	201808	255.5	292.652187	-0.126950
2	201809	225.2	233.998916	-0.037602
3	201810	204.5	190.085733	0.075830
4	201811	245.3	216.597421	0.132516
5	201812	267.3	286.408121	-0.066716
6	201901	204.4	181.692071	0.124980
7	201902	235.0	249.762622	-0.059107

```
8       201903      278.5     203.802205         0.366521
9       201904      263.5     162.345704         0.623080
10      201905      299.3     352.533799        -0.151003
11      201906      264.7     226.219424         0.170103
```

17 t-test based significance testing

[15]:
```
degreesOfFreedom = 6;

percentageDiff["tValue"] = (
    percentageDiff["percentageDiff"] / stdDev
)

from scipy.stats import t

critical_t = t.ppf(0.95, degreesOfFreedom)
critical_t

trial_results = percentageDiff[
    (percentageDiff["YEARMONTH"] >= 201902) &
    (percentageDiff["YEARMONTH"] <= 201904)
]

trial_results[["YEARMONTH", "percentageDiff", "tValue"]]
```

[15]:
```
   YEARMONTH  percentageDiff    tValue
7     201902       -0.059107 -0.593520
8     201903        0.366521  3.680430
9     201904        0.623080  6.256669
```

18 Creating Store Type + Filtering Trial & Control

[16]:
```
measureOverTimeSales = measureOverTime.copy()

measureOverTimeSales["Store_type"] = measureOverTimeSales["STORE_NBR"].apply(
    lambda x: "Trial" if x == trial_store
    else ("Control" if x == control_store else "Other")
)

pastSales = measureOverTimeSales[
    measureOverTimeSales["Store_type"].isin(["Trial", "Control"])
].copy()

pastSales["TransactionMonth"] = pd.to_datetime(
    pastSales["YEARMONTH"].astype(str) + "01",
    format="%Y%m%d"
```

```
)
pastSales
```

[16]:

	STORE_NBR	YEARMONTH	totSales	ntransactions	ncustomers	totalQty	\
880	77	201807	296.8	55	51	84	
881	77	201808	255.5	48	47	74	
882	77	201809	225.2	44	42	70	
883	77	201810	204.5	38	37	52	
884	77	201811	245.3	44	41	67	
885	77	201812	267.3	48	46	72	
886	77	201901	204.4	39	35	65	
887	77	201902	235.0	45	45	74	
888	77	201903	278.5	55	50	82	
889	77	201904	263.5	48	47	78	
890	77	201905	299.3	56	55	84	
891	77	201906	264.7	42	41	70	
2699	233	201807	290.7	54	51	88	
2700	233	201808	285.9	50	48	80	
2701	233	201809	228.6	45	42	70	
2702	233	201810	185.7	36	35	56	
2703	233	201811	211.6	41	40	62	
2704	233	201812	279.8	50	47	75	
2705	233	201901	177.5	35	35	47	
2706	233	201902	244.0	47	45	70	
2707	233	201903	199.1	41	40	59	
2708	233	201904	158.6	32	30	46	
2709	233	201905	344.4	62	57	92	
2710	233	201906	221.0	41	41	61	

	nTxnPerCust	nChipsPerTxn	avgPricePerUnit	Store_type	TransactionMonth
880	1.078431	1.527273	3.533333	Trial	2018-07-01
881	1.021277	1.541667	3.452703	Trial	2018-08-01
882	1.047619	1.590909	3.217143	Trial	2018-09-01
883	1.027027	1.368421	3.932692	Trial	2018-10-01
884	1.073171	1.522727	3.661194	Trial	2018-11-01
885	1.043478	1.500000	3.712500	Trial	2018-12-01
886	1.114286	1.666667	3.144615	Trial	2019-01-01
887	1.000000	1.644444	3.175676	Trial	2019-02-01
888	1.100000	1.490909	3.396341	Trial	2019-03-01
889	1.021277	1.625000	3.378205	Trial	2019-04-01
890	1.018182	1.500000	3.563095	Trial	2019-05-01
891	1.024390	1.666667	3.781429	Trial	2019-06-01
2699	1.058824	1.629630	3.303409	Control	2018-07-01
2700	1.041667	1.600000	3.573750	Control	2018-08-01
2701	1.071429	1.555556	3.265714	Control	2018-09-01
2702	1.028571	1.555556	3.316071	Control	2018-10-01
2703	1.025000	1.512195	3.412903	Control	2018-11-01

2704	1.063830	1.500000	3.730667	Control	2018-12-01
2705	1.000000	1.342857	3.776596	Control	2019-01-01
2706	1.044444	1.489362	3.485714	Control	2019-02-01
2707	1.025000	1.439024	3.374576	Control	2019-03-01
2708	1.066667	1.437500	3.447826	Control	2019-04-01
2709	1.087719	1.483871	3.743478	Control	2019-05-01
2710	1.000000	1.487805	3.622951	Control	2019-06-01

19 Creating 95% and 5% Confidence Bands

[17]:
```
control_sales = pastSales[pastSales["Store_type"] == "Control"].copy()

control_95 = control_sales.copy()
control_95["totSales"] = control_95["totSales"] * (1 + 2 * stdDev)
control_95["Store_type"] = "Control 95% CI"

control_5 = control_sales.copy()
control_5["totSales"] = control_5["totSales"] * (1 - 2 * stdDev)
control_5["Store_type"] = "Control 5% CI"

trialAssessment = pd.concat([pastSales, control_95, control_5])
trialAssessment
```

[17]:

	STORE_NBR	YEARMONTH	totSales	ntransactions	ncustomers	totalQty	\
880	77	201807	296.800000	55	51	84	
881	77	201808	255.500000	48	47	74	
882	77	201809	225.200000	44	42	70	
883	77	201810	204.500000	38	37	52	
884	77	201811	245.300000	44	41	67	
885	77	201812	267.300000	48	46	72	
886	77	201901	204.400000	39	35	65	
887	77	201902	235.000000	45	45	74	
888	77	201903	278.500000	55	50	82	
889	77	201904	263.500000	48	47	78	
890	77	201905	299.300000	56	55	84	
891	77	201906	264.700000	42	41	70	
2699	233	201807	290.700000	54	51	88	
2700	233	201808	285.900000	50	48	80	
2701	233	201809	228.600000	45	42	70	
2702	233	201810	185.700000	36	35	56	
2703	233	201811	211.600000	41	40	62	
2704	233	201812	279.800000	50	47	75	
2705	233	201901	177.500000	35	35	47	
2706	233	201902	244.000000	47	45	70	
2707	233	201903	199.100000	41	40	59	
2708	233	201904	158.600000	32	30	46	

2709	233	201905	344.400000	62	57	92
2710	233	201906	221.000000	41	41	61
2699	233	201807	348.599573	54	51	88
2700	233	201808	342.843543	50	48	80
2701	233	201809	274.130934	45	42	70
2702	233	201810	222.686415	36	35	56
2703	233	201811	253.744994	41	40	62
2704	233	201812	335.528588	50	47	75
2705	233	201901	212.853196	35	35	47
2706	233	201902	292.598197	47	45	70
2707	233	201903	238.755332	41	40	59
2708	233	201904	190.188828	32	30	46
2709	233	201905	412.995160	62	57	92
2710	233	201906	265.017219	41	41	61
2699	233	201807	232.800427	54	51	88
2700	233	201808	228.956457	50	48	80
2701	233	201809	183.069066	45	42	70
2702	233	201810	148.713585	36	35	56
2703	233	201811	169.455006	41	40	62
2704	233	201812	224.071412	50	47	75
2705	233	201901	142.146804	35	35	47
2706	233	201902	195.401803	47	45	70
2707	233	201903	159.444668	41	40	59
2708	233	201904	127.011172	32	30	46
2709	233	201905	275.804840	62	57	92
2710	233	201906	176.982781	41	41	61

	nTxnPerCust	nChipsPerTxn	avgPricePerUnit	Store_type	\
880	1.078431	1.527273	3.533333	Trial	
881	1.021277	1.541667	3.452703	Trial	
882	1.047619	1.590909	3.217143	Trial	
883	1.027027	1.368421	3.932692	Trial	
884	1.073171	1.522727	3.661194	Trial	
885	1.043478	1.500000	3.712500	Trial	
886	1.114286	1.666667	3.144615	Trial	
887	1.000000	1.644444	3.175676	Trial	
888	1.100000	1.490909	3.396341	Trial	
889	1.021277	1.625000	3.378205	Trial	
890	1.018182	1.500000	3.563095	Trial	
891	1.024390	1.666667	3.781429	Trial	
2699	1.058824	1.629630	3.303409	Control	
2700	1.041667	1.600000	3.573750	Control	
2701	1.071429	1.555556	3.265714	Control	
2702	1.028571	1.555556	3.316071	Control	
2703	1.025000	1.512195	3.412903	Control	
2704	1.063830	1.500000	3.730667	Control	
2705	1.000000	1.342857	3.776596	Control	

2706	1.044444	1.489362	3.485714	Control
2707	1.025000	1.439024	3.374576	Control
2708	1.066667	1.437500	3.447826	Control
2709	1.087719	1.483871	3.743478	Control
2710	1.000000	1.487805	3.622951	Control
2699	1.058824	1.629630	3.303409	Control 95% CI
2700	1.041667	1.600000	3.573750	Control 95% CI
2701	1.071429	1.555556	3.265714	Control 95% CI
2702	1.028571	1.555556	3.316071	Control 95% CI
2703	1.025000	1.512195	3.412903	Control 95% CI
2704	1.063830	1.500000	3.730667	Control 95% CI
2705	1.000000	1.342857	3.776596	Control 95% CI
2706	1.044444	1.489362	3.485714	Control 95% CI
2707	1.025000	1.439024	3.374576	Control 95% CI
2708	1.066667	1.437500	3.447826	Control 95% CI
2709	1.087719	1.483871	3.743478	Control 95% CI
2710	1.000000	1.487805	3.622951	Control 95% CI
2699	1.058824	1.629630	3.303409	Control 5% CI
2700	1.041667	1.600000	3.573750	Control 5% CI
2701	1.071429	1.555556	3.265714	Control 5% CI
2702	1.028571	1.555556	3.316071	Control 5% CI
2703	1.025000	1.512195	3.412903	Control 5% CI
2704	1.063830	1.500000	3.730667	Control 5% CI
2705	1.000000	1.342857	3.776596	Control 5% CI
2706	1.044444	1.489362	3.485714	Control 5% CI
2707	1.025000	1.439024	3.374576	Control 5% CI
2708	1.066667	1.437500	3.447826	Control 5% CI
2709	1.087719	1.483871	3.743478	Control 5% CI
2710	1.000000	1.487805	3.622951	Control 5% CI

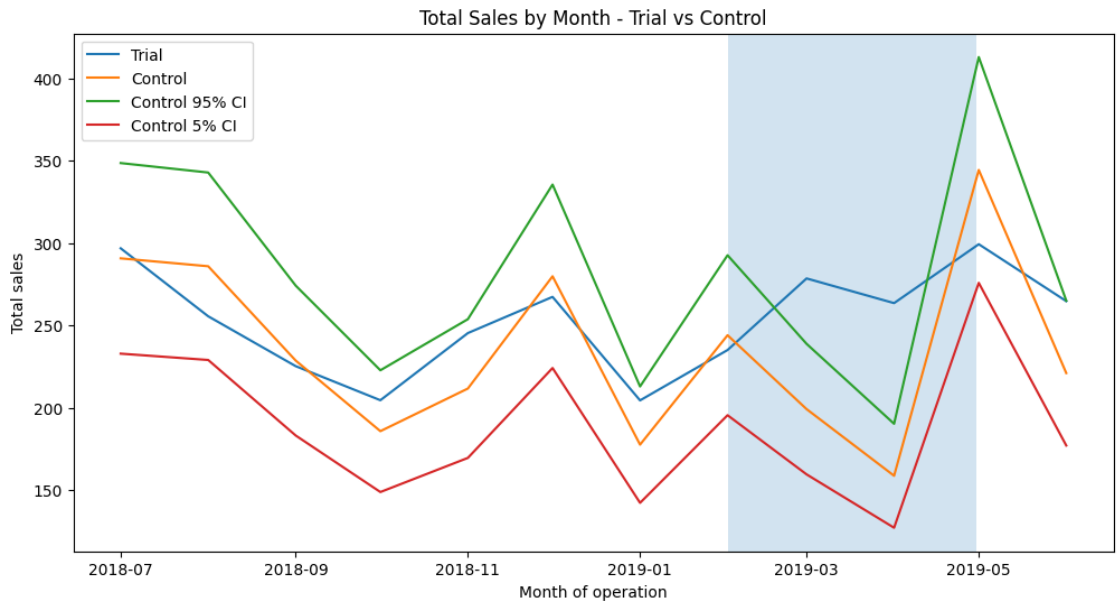
	TransactionMonth
880	2018-07-01
881	2018-08-01
882	2018-09-01
883	2018-10-01
884	2018-11-01
885	2018-12-01
886	2019-01-01
887	2019-02-01
888	2019-03-01
889	2019-04-01
890	2019-05-01
891	2019-06-01
2699	2018-07-01
2700	2018-08-01
2701	2018-09-01
2702	2018-10-01

```
2703       2018-11-01
2704       2018-12-01
2705       2019-01-01
2706       2019-02-01
2707       2019-03-01
2708       2019-04-01
2709       2019-05-01
2710       2019-06-01
2699       2018-07-01
2700       2018-08-01
2701       2018-09-01
2702       2018-10-01
2703       2018-11-01
2704       2018-12-01
2705       2019-01-01
2706       2019-02-01
2707       2019-03-01
2708       2019-04-01
2709       2019-05-01
2710       2019-06-01
2699       2018-07-01
2700       2018-08-01
2701       2018-09-01
2702       2018-10-01
2703       2018-11-01
2704       2018-12-01
2705       2019-01-01
2706       2019-02-01
2707       2019-03-01
2708       2019-04-01
2709       2019-05-01
2710       2019-06-01
```

20 Plot Everything Together

[18]:
```
plt.figure(figsize=(12,6))

for label in trialAssessment["Store_type"].unique():
    subset = trialAssessment[trialAssessment["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["totSales"], label=label)

plt.axvspan(
    pd.to_datetime("2019-02-01"),
    pd.to_datetime("2019-04-30"),
    alpha=0.2
)
```

```
plt.xlabel("Month of operation")
plt.ylabel("Total sales")
plt.title("Total Sales by Month - Trial vs Control")
plt.legend()
plt.show()
```



21 Checking if sales increase because more people visited or because customer bought more per visit

22 Computing scaling factors { Customers }

[19]:
```
trial_pre_cust = measureOverTime[
    (measureOverTime["STORE_NBR"] == trial_store) &
    (measureOverTime["YEARMONTH"] < 201902)
]["ncustomers"].sum()

control_pre_cust = measureOverTime[
    (measureOverTime["STORE_NBR"] == control_store) &
    (measureOverTime["YEARMONTH"] < 201902)
]["ncustomers"].sum()

scalingFactorForControlCust = trial_pre_cust / control_pre_cust
scalingFactorForControlCust
```

[19]: np.float64(1.0033557046979866)

23 Applying Scaling to control store

[20]:
```
measureOverTimeCusts = measureOverTime.copy()

measureOverTimeCusts["Store_type"] = measureOverTimeCusts["STORE_NBR"].apply(
    lambda x: "Trial" if x == trial_store
    else ("Control" if x == control_store else "Other")
)

scaledControlCustomers = measureOverTimeCusts[
    measureOverTimeCusts["STORE_NBR"] == control_store
].copy()

scaledControlCustomers["controlCustomers"] = (
    scaledControlCustomers["ncustomers"] * scalingFactorForControlCust
)
scaledControlCustomers
```

[20]:

	STORE_NBR	YEARMONTH	totSales	ntransactions	ncustomers	totalQty	\
2699	233	201807	290.7	54	51	88	
2700	233	201808	285.9	50	48	80	
2701	233	201809	228.6	45	42	70	
2702	233	201810	185.7	36	35	56	
2703	233	201811	211.6	41	40	62	
2704	233	201812	279.8	50	47	75	
2705	233	201901	177.5	35	35	47	
2706	233	201902	244.0	47	45	70	
2707	233	201903	199.1	41	40	59	
2708	233	201904	158.6	32	30	46	
2709	233	201905	344.4	62	57	92	
2710	233	201906	221.0	41	41	61	

	nTxnPerCust	nChipsPerTxn	avgPricePerUnit	Store_type	controlCustomers
2699	1.058824	1.629630	3.303409	Control	51.171141
2700	1.041667	1.600000	3.573750	Control	48.161074
2701	1.071429	1.555556	3.265714	Control	42.140940
2702	1.028571	1.555556	3.316071	Control	35.117450
2703	1.025000	1.512195	3.412903	Control	40.134228
2704	1.063830	1.500000	3.730667	Control	47.157718
2705	1.000000	1.342857	3.776596	Control	35.117450
2706	1.044444	1.489362	3.485714	Control	45.151007
2707	1.025000	1.439024	3.374576	Control	40.134228
2708	1.066667	1.437500	3.447826	Control	30.100671
2709	1.087719	1.483871	3.743478	Control	57.191275

```
2710     1.000000      1.487805         3.622951    Control         41.137584
```

24 Calculating Percentage Difference & Standard Deviation on pretrial only

[21]:
```
trialCustomers = measureOverTimeCusts[
    measureOverTimeCusts["STORE_NBR"] == trial_store
][["YEARMONTH", "ncustomers"]]

percentageDiff = trialCustomers.merge(
    scaledControlCustomers[["YEARMONTH", "controlCustomers"]],
    on="YEARMONTH"
)

percentageDiff["percentageDiff"] = (
    (percentageDiff["ncustomers"] - percentageDiff["controlCustomers"])
    / percentageDiff["controlCustomers"]
)
stdDev = percentageDiff[
    percentageDiff["YEARMONTH"] < 201902
]["percentageDiff"].std()


print(stdDev);
percentageDiff
```

```
0.02743996009077893
```

[21]:

	YEARMONTH	ncustomers	controlCustomers	percentageDiff
0	201807	51	51.171141	-0.003344
1	201808	47	48.161074	-0.024108
2	201809	42	42.140940	-0.003344
3	201810	37	35.117450	0.053607
4	201811	41	40.134228	0.021572
5	201812	46	47.157718	-0.024550
6	201901	35	35.117450	-0.003344
7	201902	45	45.151007	-0.003344
8	201903	50	40.134228	0.245819
9	201904	47	30.100671	0.561427
10	201905	55	57.191275	-0.038315
11	201906	41	41.137584	-0.003344

25 Creating 95% Confidence Brands

[22]:
```
degreesOfFreedom = 6

pastCustomers = measureOverTimeCusts[
    measureOverTimeCusts["Store_type"].isin(["Trial", "Control"])
].copy()

pastCustomers["TransactionMonth"] = pd.to_datetime(
    pastCustomers["YEARMONTH"].astype(str) + "01",
    format="%Y%m%d"
)

control_data = pastCustomers[
    pastCustomers["Store_type"] == "Control"
].copy()

control_95 = control_data.copy()
control_95["ncustomers"] = control_95["ncustomers"] * (1 + 2 * stdDev)
control_95["Store_type"] = "Control 95% CI"

control_5 = control_data.copy()
control_5["ncustomers"] = control_5["ncustomers"] * (1 - 2 * stdDev)
control_5["Store_type"] = "Control 5% CI"

trialAssessment = pd.concat([pastCustomers, control_95, control_5])
trialAssessment
```

[22]:

	STORE_NBR	YEARMONTH	totSales	ntransactions	ncustomers	totalQty	\
880	77	201807	296.8	55	51.000000	84	
881	77	201808	255.5	48	47.000000	74	
882	77	201809	225.2	44	42.000000	70	
883	77	201810	204.5	38	37.000000	52	
884	77	201811	245.3	44	41.000000	67	
885	77	201812	267.3	48	46.000000	72	
886	77	201901	204.4	39	35.000000	65	
887	77	201902	235.0	45	45.000000	74	
888	77	201903	278.5	55	50.000000	82	
889	77	201904	263.5	48	47.000000	78	
890	77	201905	299.3	56	55.000000	84	
891	77	201906	264.7	42	41.000000	70	
2699	233	201807	290.7	54	51.000000	88	
2700	233	201808	285.9	50	48.000000	80	
2701	233	201809	228.6	45	42.000000	70	
2702	233	201810	185.7	36	35.000000	56	
2703	233	201811	211.6	41	40.000000	62	
2704	233	201812	279.8	50	47.000000	75	

2705	233	201901	177.5	35	35.000000	47
2706	233	201902	244.0	47	45.000000	70
2707	233	201903	199.1	41	40.000000	59
2708	233	201904	158.6	32	30.000000	46
2709	233	201905	344.4	62	57.000000	92
2710	233	201906	221.0	41	41.000000	61
2699	233	201807	290.7	54	53.798876	88
2700	233	201808	285.9	50	50.634236	80
2701	233	201809	228.6	45	44.304957	70
2702	233	201810	185.7	36	36.920797	56
2703	233	201811	211.6	41	42.195197	62
2704	233	201812	279.8	50	49.579356	75
2705	233	201901	177.5	35	36.920797	47
2706	233	201902	244.0	47	47.469596	70
2707	233	201903	199.1	41	42.195197	59
2708	233	201904	158.6	32	31.646398	46
2709	233	201905	344.4	62	60.128155	92
2710	233	201906	221.0	41	43.250077	61
2699	233	201807	290.7	54	48.201124	88
2700	233	201808	285.9	50	45.365764	80
2701	233	201809	228.6	45	39.695043	70
2702	233	201810	185.7	36	33.079203	56
2703	233	201811	211.6	41	37.804803	62
2704	233	201812	279.8	50	44.420644	75
2705	233	201901	177.5	35	33.079203	47
2706	233	201902	244.0	47	42.530404	70
2707	233	201903	199.1	41	37.804803	59
2708	233	201904	158.6	32	28.353602	46
2709	233	201905	344.4	62	53.871845	92
2710	233	201906	221.0	41	38.749923	61

	nTxnPerCust	nChipsPerTxn	avgPricePerUnit	Store_type	\
880	1.078431	1.527273	3.533333	Trial	
881	1.021277	1.541667	3.452703	Trial	
882	1.047619	1.590909	3.217143	Trial	
883	1.027027	1.368421	3.932692	Trial	
884	1.073171	1.522727	3.661194	Trial	
885	1.043478	1.500000	3.712500	Trial	
886	1.114286	1.666667	3.144615	Trial	
887	1.000000	1.644444	3.175676	Trial	
888	1.100000	1.490909	3.396341	Trial	
889	1.021277	1.625000	3.378205	Trial	
890	1.018182	1.500000	3.563095	Trial	
891	1.024390	1.666667	3.781429	Trial	
2699	1.058824	1.629630	3.303409	Control	
2700	1.041667	1.600000	3.573750	Control	
2701	1.071429	1.555556	3.265714	Control	

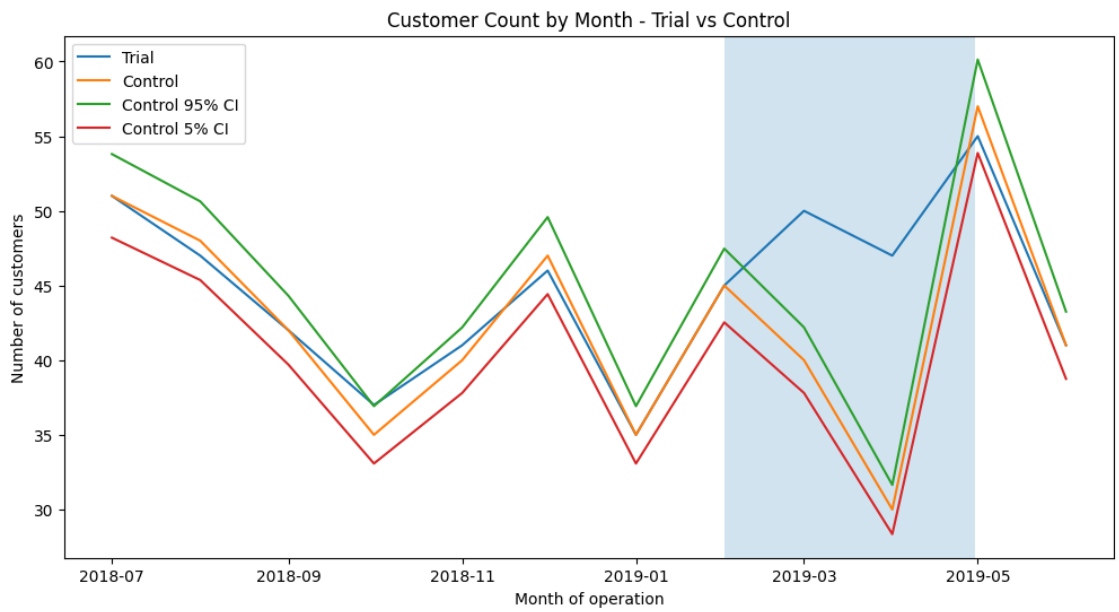
2702	1.028571	1.555556	3.316071	Control
2703	1.025000	1.512195	3.412903	Control
2704	1.063830	1.500000	3.730667	Control
2705	1.000000	1.342857	3.776596	Control
2706	1.044444	1.489362	3.485714	Control
2707	1.025000	1.439024	3.374576	Control
2708	1.066667	1.437500	3.447826	Control
2709	1.087719	1.483871	3.743478	Control
2710	1.000000	1.487805	3.622951	Control
2699	1.058824	1.629630	3.303409	Control 95% CI
2700	1.041667	1.600000	3.573750	Control 95% CI
2701	1.071429	1.555556	3.265714	Control 95% CI
2702	1.028571	1.555556	3.316071	Control 95% CI
2703	1.025000	1.512195	3.412903	Control 95% CI
2704	1.063830	1.500000	3.730667	Control 95% CI
2705	1.000000	1.342857	3.776596	Control 95% CI
2706	1.044444	1.489362	3.485714	Control 95% CI
2707	1.025000	1.439024	3.374576	Control 95% CI
2708	1.066667	1.437500	3.447826	Control 95% CI
2709	1.087719	1.483871	3.743478	Control 95% CI
2710	1.000000	1.487805	3.622951	Control 95% CI
2699	1.058824	1.629630	3.303409	Control 5% CI
2700	1.041667	1.600000	3.573750	Control 5% CI
2701	1.071429	1.555556	3.265714	Control 5% CI
2702	1.028571	1.555556	3.316071	Control 5% CI
2703	1.025000	1.512195	3.412903	Control 5% CI
2704	1.063830	1.500000	3.730667	Control 5% CI
2705	1.000000	1.342857	3.776596	Control 5% CI
2706	1.044444	1.489362	3.485714	Control 5% CI
2707	1.025000	1.439024	3.374576	Control 5% CI
2708	1.066667	1.437500	3.447826	Control 5% CI
2709	1.087719	1.483871	3.743478	Control 5% CI
2710	1.000000	1.487805	3.622951	Control 5% CI

	TransactionMonth
880	2018-07-01
881	2018-08-01
882	2018-09-01
883	2018-10-01
884	2018-11-01
885	2018-12-01
886	2019-01-01
887	2019-02-01
888	2019-03-01
889	2019-04-01
890	2019-05-01
891	2019-06-01

```
2699        2018-07-01
2700        2018-08-01
2701        2018-09-01
2702        2018-10-01
2703        2018-11-01
2704        2018-12-01
2705        2019-01-01
2706        2019-02-01
2707        2019-03-01
2708        2019-04-01
2709        2019-05-01
2710        2019-06-01
2699        2018-07-01
2700        2018-08-01
2701        2018-09-01
2702        2018-10-01
2703        2018-11-01
2704        2018-12-01
2705        2019-01-01
2706        2019-02-01
2707        2019-03-01
2708        2019-04-01
2709        2019-05-01
2710        2019-06-01
2699        2018-07-01
2700        2018-08-01
2701        2018-09-01
2702        2018-10-01
2703        2018-11-01
2704        2018-12-01
2705        2019-01-01
2706        2019-02-01
2707        2019-03-01
2708        2019-04-01
2709        2019-05-01
2710        2019-06-01
```

[23]:
```
plt.figure(figsize=(12,6))

for label in trialAssessment["Store_type"].unique():
    subset = trialAssessment[trialAssessment["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["ncustomers"], label=label)

plt.axvspan(
    pd.to_datetime("2019-02-01"),
    pd.to_datetime("2019-04-30"),
    alpha=0.2
```

```
)

plt.xlabel("Month of operation")
plt.ylabel("Number of customers")
plt.title("Customer Count by Month - Trial vs Control")
plt.legend()
plt.show()
```



25.0.1 Hence, in Trial period (Feb–Apr), Sales/customers increased significantly & After trial numbers revert, indicating the effect was driven by the trial itself, not seasonal trends

26 For trial store 86

27 Calculating and combining Correlation & Magnitude

[24]:
```
trial_store = 86;

#Correlation
corr_nSales = calculate_correlation(preTrialMeasures,"totSales",trial_store) ;
corr_nCustomers =␣
 ↪calculate_correlation(preTrialMeasures,"ncustomers",trial_store) ;

#Magnitude
magnitude_nSales =␣
 ↪calculate_magnitude_distance(preTrialMeasures,"totSales",trial_store) ;
```

```
magnitude_nCustomers =␣
↪calculate_magnitude_distance(preTrialMeasures,"ncustomers",trial_store) ;

corr_weight = 0.5

score_nSales = (corr_nSales.merge(magnitude_nSales, on=["Store1", "Store2"]));

score_nSales["scoreNSales"] = (corr_weight * score_nSales["corr_measure"] + (1␣
↪- corr_weight) * score_nSales["mag_measure"]);

score_nCustomers = (corr_nCustomers.merge(magnitude_nCustomers, on=["Store1",␣
↪"Store2"]));

score_nCustomers["scoreNCust"] = (corr_weight *␣
↪score_nCustomers["corr_measure"] +(1 - corr_weight) *␣
↪score_nCustomers["mag_measure"]);
```

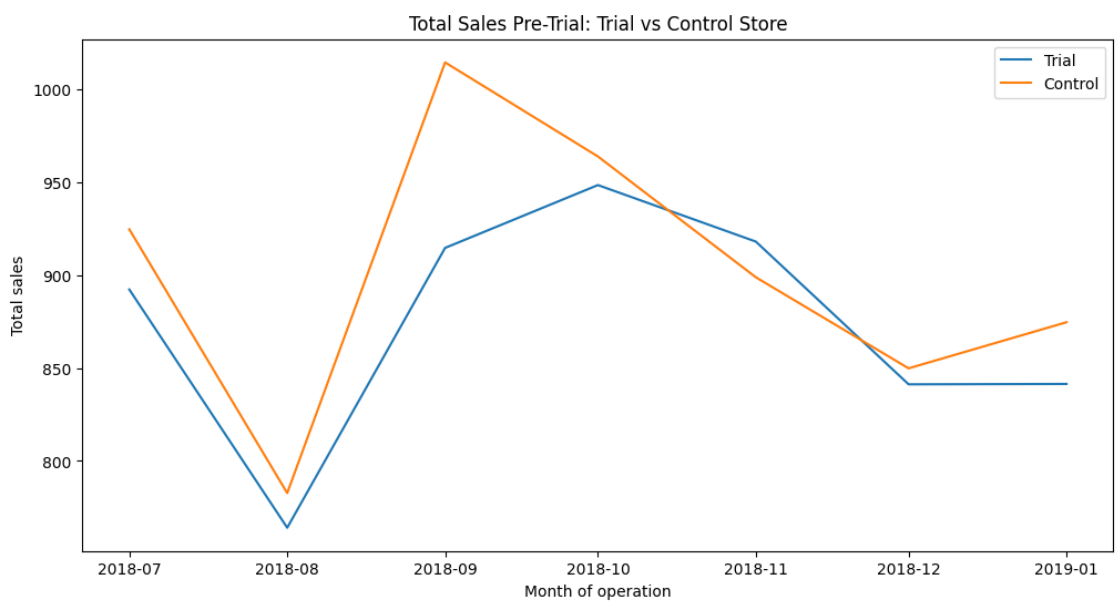
28 Combine Drivers (Simple Average) & Selecting best control store

[25]:
```
score_Control = (
    score_nSales[["Store1", "Store2", "scoreNSales"]]
    .merge(
        score_nCustomers[["Store1", "Store2", "scoreNCust"]],
        on=["Store1", "Store2"]
    )
)

score_Control["finalControlScore"] = (
    0.5 * score_Control["scoreNSales"] +
    0.5 * score_Control["scoreNCust"]
)

control_store = (
    score_Control
    .sort_values("finalControlScore", ascending=False)
    .iloc[0]["Store2"]
)

control_store
```

[25]: np.float64(155.0)

29 Visual Check of total Sales

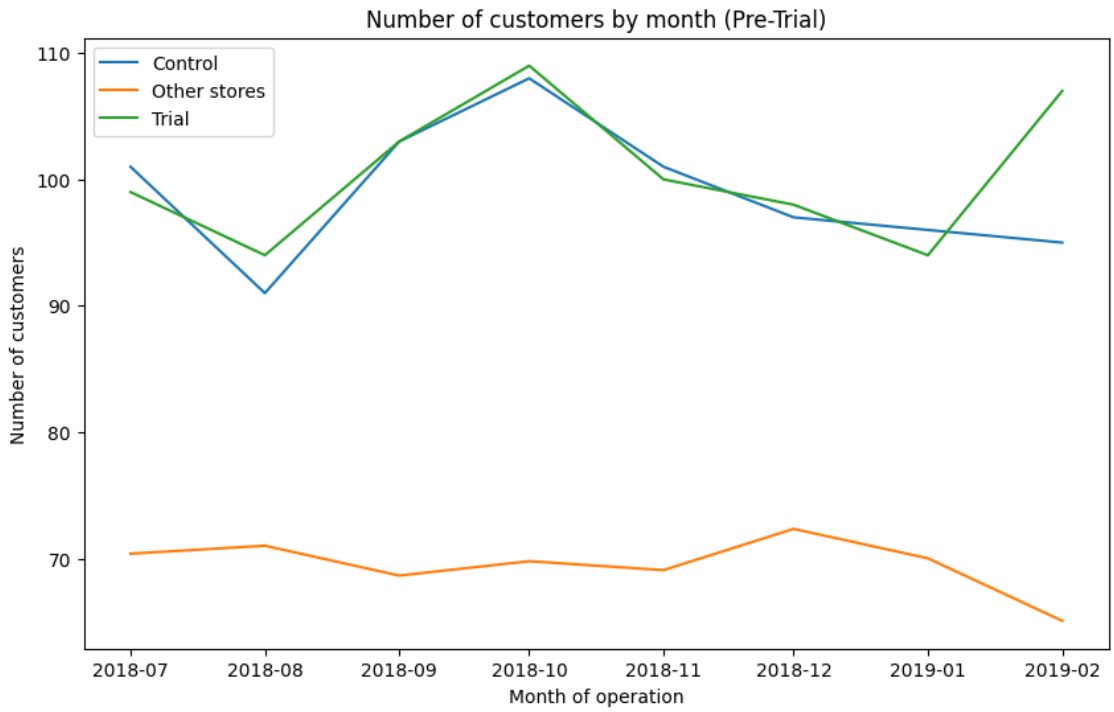
[26]:
```
measureOverTimeSales = measureOverTime.copy()

measureOverTimeSales["Store_type"] = measureOverTimeSales["STORE_NBR"].apply(
    lambda x: "Trial" if x == trial_store else ("Control" if x == control_store␣
 ↪else "Other")
)

pastSales = measureOverTimeSales.groupby(
    ["YEARMONTH", "Store_type"], as_index=False
)["totSales"].mean()

pastSales["TransactionMonth"] = pd.to_datetime(
    pastSales["YEARMONTH"].astype(str) + "01", format="%Y%m%d"
)

pastSales = pastSales[pastSales["YEARMONTH"] < 201902]


plt.figure(figsize=(12,6))
for label in ["Trial", "Control"]:
    subset = pastSales[pastSales["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["totSales"], label=label)

plt.xlabel("Month of operation")
plt.ylabel("Total sales")
plt.title("Total Sales Pre-Trial: Trial vs Control Store")
plt.legend()
plt.show()
```



29.0.1 Great, sales are trending in a similar way.

30 Visual check of no of customers

[27]:
```
measureOverTimeCusts = measureOverTime.copy()

measureOverTimeCusts["Store_type"] = np.where(
    measureOverTimeCusts["STORE_NBR"] == trial_store, "Trial",
    np.where(measureOverTimeCusts["STORE_NBR"] == control_store, "Control",␣
 ↪"Other stores")
)

pastCustomers = measureOverTimeCusts.groupby(["YEARMONTH",␣
 ↪"Store_type"])["ncustomers"].mean().reset_index()

pastCustomers["TransactionMonth"] = pd.to_datetime(pastCustomers["YEARMONTH"].
 ↪astype(str) + "01", format="%Y%m%d")

pastCustomers = pastCustomers[pastCustomers["YEARMONTH"] < 201903]

plt.figure(figsize=(10, 6))
for label in pastCustomers["Store_type"].unique():
    subset = pastCustomers[pastCustomers["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["ncustomers"], label=label)
plt.xlabel("Month of operation")
plt.ylabel("Number of customers")
plt.title("Number of customers by month (Pre-Trial)")
plt.legend()
plt.show()
```



30.0.1 Good, the trend in number of customers is also similar.

31 Checking if sales significantly increase during Feb–Apr 2019

32 Computing Scaling Factor {Sales}

[28]:
```
scalingFactorForControlSales = (
    preTrialMeasures[preTrialMeasures["STORE_NBR"] == trial_store]["totSales"].
 ↪sum() /
    preTrialMeasures[preTrialMeasures["STORE_NBR"] ==␣
 ↪control_store]["totSales"].sum()
)

scaledControlSales = measureOverTime[measureOverTime["STORE_NBR"] ==␣
 ↪control_store].copy()
scaledControlSales["controlSales"] = scaledControlSales["totSales"] *␣
 ↪scalingFactorForControlSales

percentageDiff = (
    measureOverTime[measureOverTime["STORE_NBR"] == trial_store][["YEARMONTH",␣
 ↪"totSales"]]
    .merge(scaledControlSales[["YEARMONTH", "controlSales"]], on="YEARMONTH")
)
```

```
percentageDiff["percentageDiff"] = (
    (percentageDiff["totSales"] - percentageDiff["controlSales"]) /␣
 ↪percentageDiff["controlSales"]
)

stdDev = percentageDiff[percentageDiff["YEARMONTH"] < 201902]["percentageDiff"].
 ↪std()

degreesOfFreedom = 7

percentageDiff["tValue"] = percentageDiff["percentageDiff"] / stdDev
percentageDiff
```

[28]:

	YEARMONTH	totSales	controlSales	percentageDiff	tValue
0	201807	892.20	896.922236	-0.005265	-0.139708
1	201808	764.05	759.269991	0.006296	0.167055
2	201809	914.60	984.034086	-0.070561	-1.872364
3	201810	948.40	934.948790	0.014387	0.381770
4	201811	918.00	871.894555	0.052880	1.403188
5	201812	841.20	824.361363	0.020426	0.542022
6	201901	841.40	848.418979	-0.008273	-0.219529
7	201902	913.20	864.522060	0.056306	1.494114
8	201903	1026.80	780.320405	0.315870	8.381769
9	201904	848.20	819.317024	0.035253	0.935444
10	201905	889.30	895.224622	-0.006618	-0.175613
11	201906	838.00	831.539845	0.007769	0.206152

33 Combine trial assessment with pre-trial, trial, and post-trial data for Total Sales

[29]:

```
measureOverTimeSales = measureOverTime.copy()
measureOverTimeSales["TransactionMonth"] = pd.to_datetime(
    measureOverTimeSales["YEARMONTH"].astype(str) + "01", format="%Y%m%d"
)

trial_sales = measureOverTimeSales[measureOverTimeSales["STORE_NBR"] ==␣
 ↪trial_store].copy()
trial_sales["Store_type"] = "Trial"

control_sales = measureOverTimeSales[measureOverTimeSales["STORE_NBR"] ==␣
 ↪control_store].copy()
control_sales["Store_type"] = "Control"

scaled_control_sales = control_sales.copy()
scaled_control_sales["totSales"] = scaled_control_sales["totSales"] *␣
 ↪scalingFactorForControlSales
```

```
scaled_control_sales["Store_type"] = "Control"

trialAssessmentSales = pd.concat([trial_sales, scaled_control_sales])

control_95_sales = control_sales.copy()
control_95_sales["totSales"] = control_95_sales["totSales"] *␣
↪scalingFactorForControlSales * (1 + stdDev * 2)
control_95_sales["Store_type"] = "Control 95% CI"

control_5_sales = control_sales.copy()
control_5_sales["totSales"] = control_5_sales["totSales"] *␣
↪scalingFactorForControlSales * (1 - stdDev * 2)
control_5_sales["Store_type"] = "Control 5% CI"

trialAssessmentSales = pd.concat([trialAssessmentSales, control_95_sales,␣
↪control_5_sales])
trialAssessmentSales.sort_values(["YEARMONTH", "Store_type"]).head()
```

[29]:

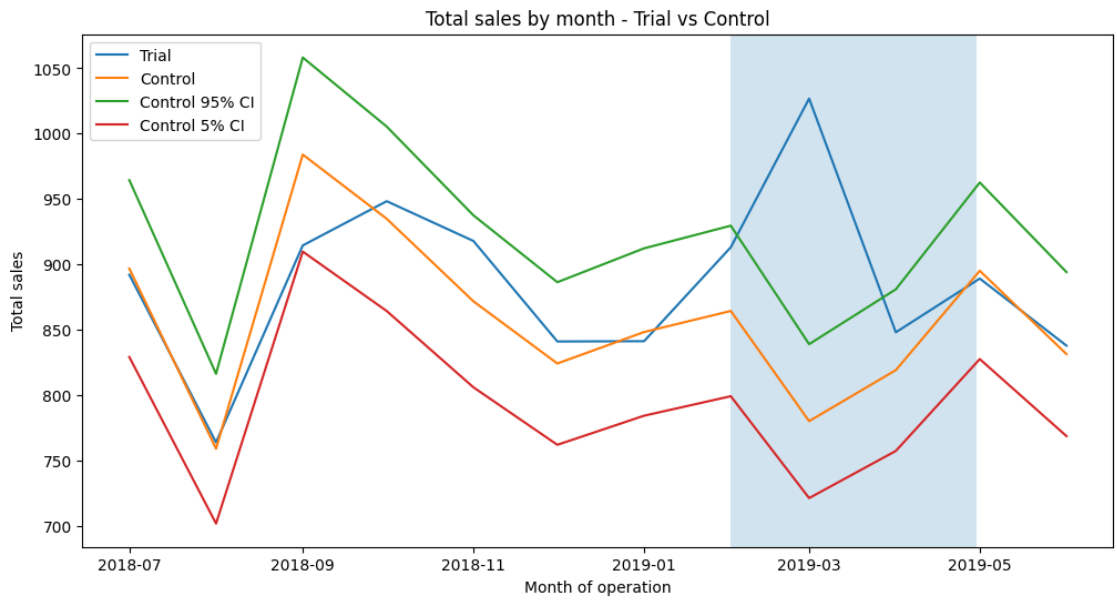
```
      STORE_NBR  YEARMONTH    totSales  ntransactions  ncustomers  totalQty  \
1793        155     201807  896.922236            123         101       250
1793        155     201807  829.320619            123         101       250
1793        155     201807  964.523853            123         101       250
977          86     201807  892.200000            126          99       251
1794        155     201808  759.269991            119          91       229

      nTxnPerCust  nChipsPerTxn  avgPricePerUnit TransactionMonth  \
1793     1.217822      2.032520         3.698400       2018-07-01
1793     1.217822      2.032520         3.698400       2018-07-01
1793     1.217822      2.032520         3.698400       2018-07-01
977      1.272727      1.992063         3.554582       2018-07-01
1794     1.307692      1.924370         3.417904       2018-08-01

          Store_type
1793         Control
1793   Control 5% CI
1793  Control 95% CI
977            Trial
1794         Control
```

34 Plotting Everything Together

[30]:

```
plt.figure(figsize=(12,6))
for label in trialAssessmentSales["Store_type"].unique():
    subset = trialAssessmentSales[trialAssessmentSales["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["totSales"], label=label)
```

```
plt.axvspan(
    pd.to_datetime("2019-02-01"),
    pd.to_datetime("2019-04-30"),
    alpha=0.2
)
plt.xlabel("Month of operation")
plt.ylabel("Total sales")
plt.title("Total sales by month - Trial vs Control")
plt.legend()
plt.show()
```



35 Checking if sales increase because more people visited or because customer bought more per visit

36 Computing Scaling Factor {Customers}

[31]:
```
scalingFactorForControlCust = (
    preTrialMeasures[preTrialMeasures["STORE_NBR"] ==␣
 ↪trial_store]["ncustomers"].sum() /
    preTrialMeasures[preTrialMeasures["STORE_NBR"] ==␣
 ↪control_store]["ncustomers"].sum()
)

scaledControlCustomers = measureOverTime[measureOverTime["STORE_NBR"] ==␣
 ↪control_store].copy()
```

```
scaledControlCustomers["controlCustomers"] =␣
↪scaledControlCustomers["ncustomers"] * scalingFactorForControlCust

percentageDiffCust = (
    measureOverTime[measureOverTime["STORE_NBR"] == trial_store][["YEARMONTH",␣
↪"ncustomers"]]
    .merge(scaledControlCustomers[["YEARMONTH", "controlCustomers"]],␣
↪on="YEARMONTH")
)
percentageDiffCust["percentageDiff"] = (
    (percentageDiffCust["ncustomers"] - percentageDiffCust["controlCustomers"])␣
↪/ percentageDiffCust["controlCustomers"]
)

stdDevCust = percentageDiffCust[percentageDiffCust["YEARMONTH"] <␣
↪201902]["percentageDiff"].std()

percentageDiffCust["tValue"] = percentageDiffCust["percentageDiff"] /␣
↪stdDevCust;
percentageDiffCust
```

[31]:

	YEARMONTH	ncustomers	controlCustomers	percentageDiff	tValue
0	201807	99	101.0	-0.019802	-1.033550
1	201808	94	91.0	0.032967	1.720691
2	201809	103	103.0	0.000000	0.000000
3	201810	109	108.0	0.009259	0.483281
4	201811	100	101.0	-0.009901	-0.516775
5	201812	98	97.0	0.010309	0.538086
6	201901	94	96.0	-0.020833	-1.087381
7	201902	107	95.0	0.126316	6.592964
8	201903	115	94.0	0.223404	11.660429
9	201904	105	99.0	0.060606	3.163291
10	201905	104	106.0	-0.018868	-0.984798
11	201906	98	95.0	0.031579	1.648241

37 Combine trial assessment with pre-trial, trial, and post-trial data for Number of Customers

[32]:

```
measureOverTimeCusts = measureOverTime.copy()
measureOverTimeCusts["TransactionMonth"] = pd.to_datetime(
    measureOverTimeCusts["YEARMONTH"].astype(str) + "01", format="%Y%m%d"
)

trial_cust = measureOverTimeCusts[measureOverTimeCusts["STORE_NBR"] ==␣
↪trial_store].copy()
trial_cust["Store_type"] = "Trial"
```

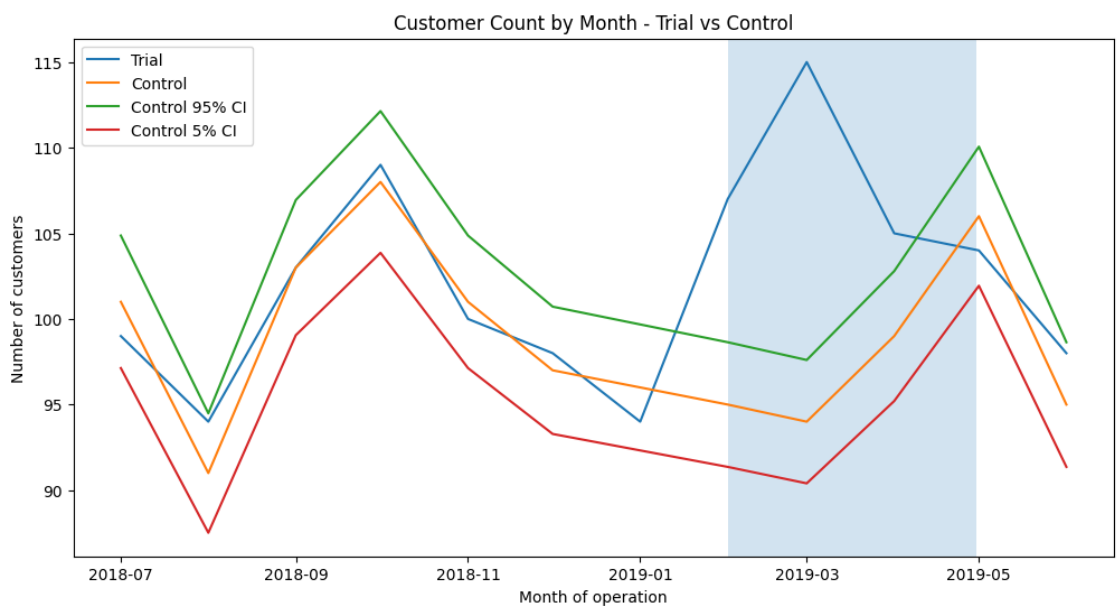
```
control_cust = measureOverTimeCusts[measureOverTimeCusts["STORE_NBR"] ==␣
↪control_store].copy()
control_cust["Store_type"] = "Control"

scaled_control_cust = control_cust.copy()
scaled_control_cust["ncustomers"] = scaled_control_cust["ncustomers"] *␣
↪scalingFactorForControlCust
scaled_control_cust["Store_type"] = "Control"

trialAssessmentCust = pd.concat([trial_cust, scaled_control_cust])

control_95_cust = control_cust.copy()
control_95_cust["ncustomers"] = control_95_cust["ncustomers"] *␣
↪scalingFactorForControlCust * (1 + stdDevCust * 2)
control_95_cust["Store_type"] = "Control 95% CI"

control_5_cust = control_cust.copy()
control_5_cust["ncustomers"] = control_5_cust["ncustomers"] *␣
↪scalingFactorForControlCust * (1 - stdDevCust * 2)
control_5_cust["Store_type"] = "Control 5% CI"

trialAssessmentCust = pd.concat([trialAssessmentCust, control_95_cust,␣
↪control_5_cust])
trialAssessmentCust.sort_values(["YEARMONTH", "Store_type"]).head()
```

[32]:

```
      STORE_NBR  YEARMONTH  totSales  ntransactions  ncustomers  totalQty  \
1793        155     201807     924.6            123  101.000000       250
1793        155     201807     924.6            123   97.129846       250
1793        155     201807     924.6            123  104.870154       250
977          86     201807     892.2            126   99.000000       251
1794        155     201808     782.7            119   91.000000       229

      nTxnPerCust  nChipsPerTxn  avgPricePerUnit TransactionMonth  \
1793     1.217822      2.032520         3.698400       2018-07-01
1793     1.217822      2.032520         3.698400       2018-07-01
1793     1.217822      2.032520         3.698400       2018-07-01
977      1.272727      1.992063         3.554582       2018-07-01
1794     1.307692      1.924370         3.417904       2018-08-01

          Store_type
1793         Control
1793   Control 5% CI
1793  Control 95% CI
977            Trial
1794         Control
```

38 Plotting Everything Together

[33]:
```
plt.figure(figsize=(12,6))
for label in trialAssessmentCust["Store_type"].unique():
    subset = trialAssessmentCust[trialAssessmentCust["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["ncustomers"], label=label)

plt.axvspan(
    pd.to_datetime("2019-02-01"),
    pd.to_datetime("2019-04-30"),
    alpha=0.2
)
plt.xlabel("Month of operation")
plt.ylabel("Number of customers")
plt.title("Customer Count by Month - Trial vs Control")
plt.legend()
plt.show()
```



38.0.1 Hence, according to the plots and the way the trial store behaves relative to the scaled control store + confidence intervals, there is a clear and significant uplift during the trial period (Feb–Apr 2019) for store 86 compared to its matched control store 155.

39 For trial store 88

40 Calculating and combining Correlation & Magnitude

[34]:
```
trial_store = 88

# Correlation
corr_nSales    = calculate_correlation(preTrialMeasures, "totSales",   ␣
 ↪trial_store)
corr_nCustomers = calculate_correlation(preTrialMeasures, "ncustomers",␣
 ↪trial_store)

# Magnitude distance
magnitude_nSales    = calculate_magnitude_distance(preTrialMeasures,␣
 ↪"totSales",    trial_store)
magnitude_nCustomers = calculate_magnitude_distance(preTrialMeasures,␣
 ↪"ncustomers", trial_store)

corr_weight = 0.5

# Combine correlation & magnitude
score_nSales = corr_nSales.merge(magnitude_nSales, on=["Store1", "Store2"])
score_nSales["scoreNSales"] = (
    corr_weight * score_nSales["corr_measure"] +
    (1 - corr_weight) * score_nSales["mag_measure"]
)

score_nCustomers = corr_nCustomers.merge(magnitude_nCustomers, on=["Store1",␣
 ↪"Store2"])
score_nCustomers["scoreNCust"] = (
    corr_weight * score_nCustomers["corr_measure"] +
    (1 - corr_weight) * score_nCustomers["mag_measure"]
)
score_nCustomers
```

[34]:

	Store1	Store2	corr_measure	mag_measure	scoreNCust
0	88	1	0.305334	0.357143	0.331238
1	88	2	-0.452379	0.285714	-0.083332
2	88	3	0.522884	0.683673	0.603279
3	88	4	-0.361503	0.577922	0.108210
4	88	5	-0.025320	0.558442	0.266561
..	…	…	…	…	…

```
254      88     268      0.672672     0.630252    0.651462
255      88     269     -0.274781     0.360902    0.043061
256      88     270     -0.103032     0.367965    0.132467
257      88     271     -0.018831     0.655844    0.318507
258      88     272      0.026909     0.448980    0.237944

[259 rows x 5 columns]
```

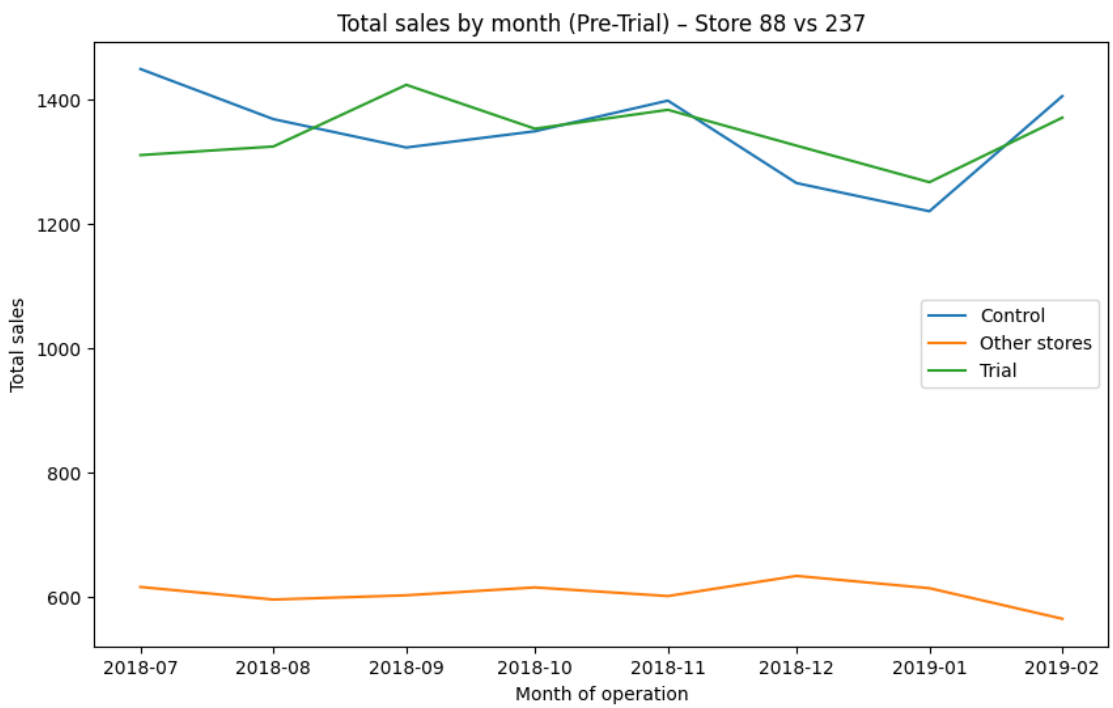
41 Combine Drivers (Simple Average) & Selecting best control store

[35]:
```
score_Control = (
    score_nSales[["Store1", "Store2", "scoreNSales"]]
    .merge(
        score_nCustomers[["Store1", "Store2", "scoreNCust"]],
        on=["Store1", "Store2"]
    )
)

score_Control["finalControlScore"] = (
    0.5 * score_Control["scoreNSales"] +
    0.5 * score_Control["scoreNCust"]
)

control_store = (
    score_Control
    .sort_values("finalControlScore", ascending=False)
    .iloc[0]["Store2"]
)

score_Control.sort_values("finalControlScore", ascending=False)

control_store = 237 ;
```

42 Visual checks on trends based on the drivers – Pre-Trial Total Sales

[36]:
```
measureOverTimeSales = measureOverTime.copy()

measureOverTimeSales["Store_type"] = np.where(
    measureOverTimeSales["STORE_NBR"] == trial_store, "Trial",
    np.where(measureOverTimeSales["STORE_NBR"] == control_store, "Control",␣
 ↪"Other stores")
)
```

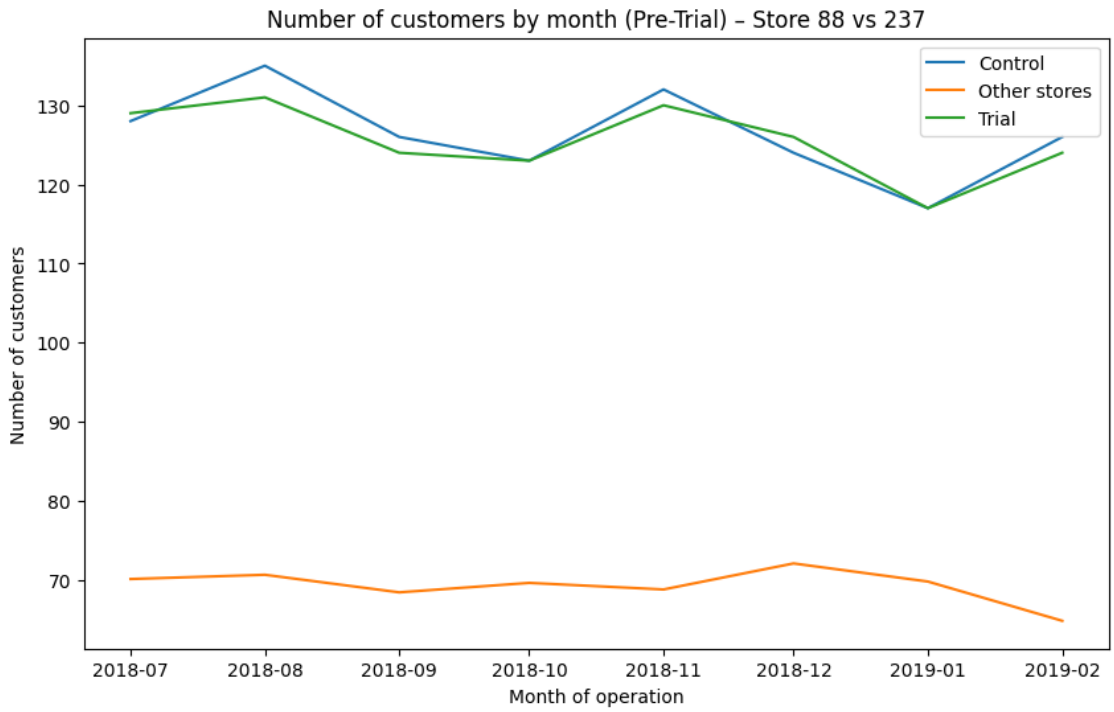
```
pastSales = measureOverTimeSales.groupby(["YEARMONTH",␣
↪"Store_type"])["totSales"].mean().reset_index()

pastSales["TransactionMonth"] = pd.to_datetime(
    pastSales["YEARMONTH"].astype(str) + "01", format="%Y%m%d"
)

pastSales = pastSales[pastSales["YEARMONTH"] < 201903]

plt.figure(figsize=(10, 6))
for label in pastSales["Store_type"].unique():
    subset = pastSales[pastSales["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["totSales"], label=label)

plt.xlabel("Month of operation")
plt.ylabel("Total sales")
plt.title("Total sales by month (Pre-Trial) - Store 88 vs 237")
plt.legend()
plt.show()
```



42.0.1 the trial and control stores have similar total sales trends before the trial.

43 Visual checks on trends based on the drivers – Pre-Trial Number of Customers

[37]:
```
measureOverTimeCusts = measureOverTime.copy()

measureOverTimeCusts["Store_type"] = np.where(
    measureOverTimeCusts["STORE_NBR"] == trial_store, "Trial",
    np.where(measureOverTimeCusts["STORE_NBR"] == control_store, "Control",␣
 ↪"Other stores")
)

pastCustomers = measureOverTimeCusts.groupby(["YEARMONTH",␣
 ↪"Store_type"])["ncustomers"].mean().reset_index()

pastCustomers["TransactionMonth"] = pd.to_datetime(
    pastCustomers["YEARMONTH"].astype(str) + "01", format="%Y%m%d"
)

pastCustomers = pastCustomers[pastCustomers["YEARMONTH"] < 201903]

plt.figure(figsize=(10, 6))
for label in pastCustomers["Store_type"].unique():
    subset = pastCustomers[pastCustomers["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["ncustomers"], label=label)

plt.xlabel("Month of operation")
plt.ylabel("Number of customers")
plt.title("Number of customers by month (Pre-Trial) - Store 88 vs 237")
plt.legend()
plt.show()
```



43.0.1 otal number of customers of the control and trial stores are also similar.

44 Calculating Scaling Factors for sale.

[38]:
```
scalingFactorForControlSales = (
    preTrialMeasures[preTrialMeasures["STORE_NBR"] == trial_store]["totSales"].
 ↪sum() /
    preTrialMeasures[preTrialMeasures["STORE_NBR"] ==␣
 ↪control_store]["totSales"].sum()
)

scaledControlSales = measureOverTime[measureOverTime["STORE_NBR"] ==␣
 ↪control_store].copy()
scaledControlSales["controlSales"] = scaledControlSales["totSales"] *␣
 ↪scalingFactorForControlSales

percentageDiff = (
    measureOverTime[measureOverTime["STORE_NBR"] == trial_store][["YEARMONTH",␣
 ↪"totSales"]]
    .merge(scaledControlSales[["YEARMONTH", "controlSales"]], on="YEARMONTH")
)
percentageDiff["percentageDiff"] = (
```

```
    (percentageDiff["totSales"] - percentageDiff["controlSales"]) /␣
 ↪percentageDiff["controlSales"]
)

stdDev = percentageDiff[percentageDiff["YEARMONTH"] < 201902]["percentageDiff"].
 ↪std()
percentageDiff.head()
```

[38]:

	YEARMONTH	totSales	controlSales	percentageDiff
0	201807	1310.0	1450.657086	-0.096961
1	201808	1323.8	1369.931485	-0.033674
2	201809	1423.0	1324.260425	0.074562
3	201810	1352.4	1350.401097	0.001480
4	201811	1382.8	1399.777923	-0.012129

45 Combine trial assessment with pre-trial, trial, and post-trial data for Total Sales

[39]:

```
measureOverTimeSales = measureOverTime.copy()
measureOverTimeSales["TransactionMonth"] = pd.to_datetime(
    measureOverTimeSales["YEARMONTH"].astype(str) + "01", format="%Y%m%d"
)

trial_sales = measureOverTimeSales[measureOverTimeSales["STORE_NBR"] ==␣
 ↪trial_store].copy()
trial_sales["Store_type"] = "Trial"

control_sales = measureOverTimeSales[measureOverTimeSales["STORE_NBR"] ==␣
 ↪control_store].copy()
control_sales["Store_type"] = "Control"

scaled_control_sales = control_sales.copy()
scaled_control_sales["totSales"] = scaled_control_sales["totSales"] *␣
 ↪scalingFactorForControlSales
scaled_control_sales["Store_type"] = "Control"

trialAssessment = pd.concat([trial_sales, scaled_control_sales])

control_95 = control_sales.copy()
control_95["totSales"] = control_95["totSales"] * scalingFactorForControlSales␣
 ↪* (1 + stdDev * 2)
control_95["Store_type"] = "Control 95% CI"

control_5 = control_sales.copy()
control_5["totSales"] = control_5["totSales"] * scalingFactorForControlSales *␣
 ↪(1 - stdDev * 2)
```

```
control_5["Store_type"] = "Control 5% CI"

trialAssessment = pd.concat([trialAssessment, control_95, control_5])
trialAssessment.head()
```

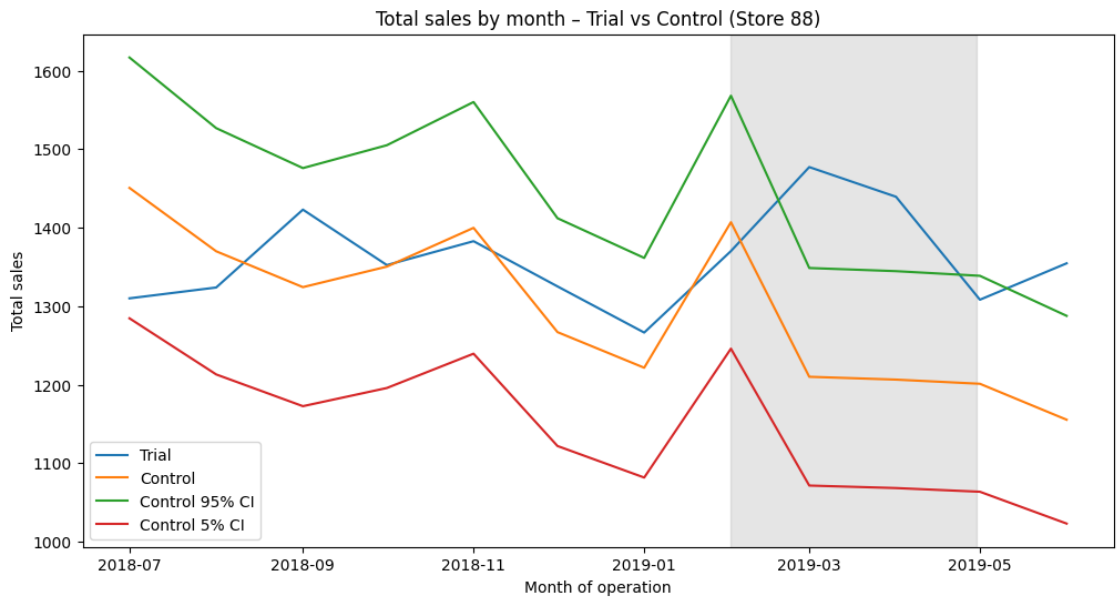
[39]:

```
      STORE_NBR  YEARMONTH  totSales  ntransactions  ncustomers  totalQty  \
1001         88     201807    1310.0            153         129       306
1002         88     201808    1323.8            158         131       303
1003         88     201809    1423.0            157         124       318
1004         88     201810    1352.4            155         123       316
1005         88     201811    1382.8            156         130       314

      nTxnPerCust  nChipsPerTxn  avgPricePerUnit TransactionMonth Store_type
1001     1.186047      2.000000         4.281046       2018-07-01      Trial
1002     1.206107      1.917722         4.368977       2018-08-01      Trial
1003     1.266129      2.025478         4.474843       2018-09-01      Trial
1004     1.260163      2.038710         4.279747       2018-10-01      Trial
1005     1.200000      2.012821         4.403822       2018-11-01      Trial
```

46 Plotting Everything Together

[40]:

```
plt.figure(figsize=(12,6))
for label in trialAssessment["Store_type"].unique():
    subset = trialAssessment[trialAssessment["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["totSales"], label=label)

plt.axvspan(
    pd.to_datetime("2019-02-01"),
    pd.to_datetime("2019-04-30"),
    alpha=0.2, color='gray'
)
plt.xlabel("Month of operation")
plt.ylabel("Total sales")
plt.title("Total sales by month - Trial vs Control (Store 88)")
plt.legend()
plt.show()
```



47 Calculating Scaling Factors for no of customers.

[41]:
```
scalingFactorForControlCust = (
    preTrialMeasures[preTrialMeasures["STORE_NBR"] ==␣
 ↪trial_store]["ncustomers"].sum() /
    preTrialMeasures[preTrialMeasures["STORE_NBR"] ==␣
 ↪control_store]["ncustomers"].sum()
)

scaledControlCustomers = measureOverTime[measureOverTime["STORE_NBR"] ==␣
 ↪control_store].copy()
scaledControlCustomers["controlCustomers"] = (
    scaledControlCustomers["ncustomers"] * scalingFactorForControlCust
)

percentageDiffCust = (
    measureOverTime[measureOverTime["STORE_NBR"] == trial_store][["YEARMONTH",␣
 ↪"ncustomers"]]
    .merge(scaledControlCustomers[["YEARMONTH", "controlCustomers"]],␣
 ↪on="YEARMONTH")
)
percentageDiffCust["percentageDiff"] = (
    (percentageDiffCust["ncustomers"] - percentageDiffCust["controlCustomers"])␣
 ↪/
    percentageDiffCust["controlCustomers"]
)
```

```
stdDevCust = percentageDiffCust[percentageDiffCust["YEARMONTH"] <␣
↪201902]["percentageDiff"].std()
percentageDiffCust.head()
```

[41]:

	YEARMONTH	ncustomers	controlCustomers	percentageDiff
0	201807	129	127.276836	0.013539
1	201808	131	134.237288	-0.024116
2	201809	124	125.288136	-0.010281
3	201810	123	122.305085	0.005682
4	201811	130	131.254237	-0.009556

48 Combine trial assessment with pre-trial, trial, and post-trial data for Customers

[42]:

```
measureOverTimeCusts = measureOverTime.copy()
measureOverTimeCusts["TransactionMonth"] = pd.to_datetime(
    measureOverTimeCusts["YEARMONTH"].astype(str) + "01", format="%Y%m%d"
)

trial_cust = measureOverTimeCusts[measureOverTimeCusts["STORE_NBR"] ==␣
↪trial_store].copy()
trial_cust["Store_type"] = "Trial"

control_cust = measureOverTimeCusts[measureOverTimeCusts["STORE_NBR"] ==␣
↪control_store].copy()
control_cust["Store_type"] = "Control"

scaled_control_cust = control_cust.copy()
scaled_control_cust["ncustomers"] = scaled_control_cust["ncustomers"] *␣
↪scalingFactorForControlCust
scaled_control_cust["Store_type"] = "Control"

trialAssessmentCust = pd.concat([trial_cust, scaled_control_cust])

control_95_cust = control_cust.copy()
control_95_cust["ncustomers"] = control_95_cust["ncustomers"] *␣
↪scalingFactorForControlCust * (1 + stdDevCust * 2)
control_95_cust["Store_type"] = "Control 95% CI"

control_5_cust = control_cust.copy()
control_5_cust["ncustomers"] = control_5_cust["ncustomers"] *␣
↪scalingFactorForControlCust * (1 - stdDevCust * 2)
control_5_cust["Store_type"] = "Control 5% CI"
```

```
trialAssessmentCust = pd.concat([trialAssessmentCust, control_95_cust,␣
↪control_5_cust])

trialAssessmentCust.head()
```

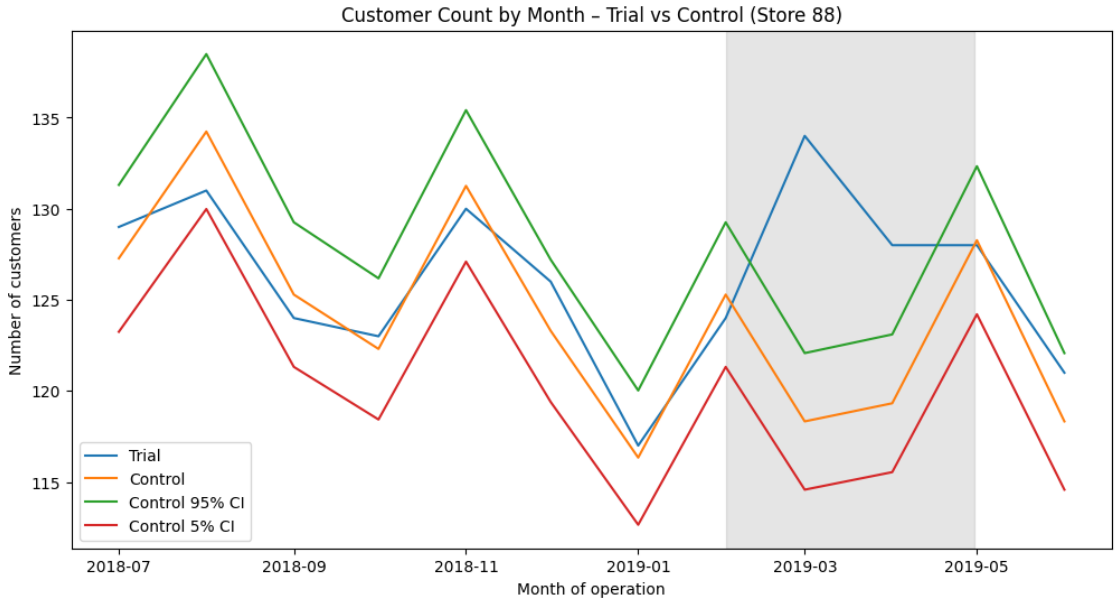
[42]:

```
      STORE_NBR  YEARMONTH  totSales  ntransactions  ncustomers  totalQty  \
1001         88     201807    1310.0            153       129.0       306
1002         88     201808    1323.8            158       131.0       303
1003         88     201809    1423.0            157       124.0       318
1004         88     201810    1352.4            155       123.0       316
1005         88     201811    1382.8            156       130.0       314

      nTxnPerCust  nChipsPerTxn  avgPricePerUnit TransactionMonth Store_type
1001     1.186047      2.000000         4.281046       2018-07-01      Trial
1002     1.206107      1.917722         4.368977       2018-08-01      Trial
1003     1.266129      2.025478         4.474843       2018-09-01      Trial
1004     1.260163      2.038710         4.279747       2018-10-01      Trial
1005     1.200000      2.012821         4.403822       2018-11-01      Trial
```

49 Plotting Everything together

[43]:

```
plt.figure(figsize=(12,6))
for label in trialAssessmentCust["Store_type"].unique():
    subset = trialAssessmentCust[trialAssessmentCust["Store_type"] == label]
    plt.plot(subset["TransactionMonth"], subset["ncustomers"], label=label)

plt.axvspan(
    pd.to_datetime("2019-02-01"),
    pd.to_datetime("2019-04-30"),
    alpha=0.2, color='gray'
)
plt.xlabel("Month of operation")
plt.ylabel("Number of customers")
plt.title("Customer Count by Month - Trial vs Control (Store 88)")
plt.legend()
plt.show()
```



49.0.1 Total number of customers in the trial period for the trial store is significantly higher than the control store for two out of three months

50 Conclusion

We've found control stores 233, 155, 237 for trial stores 77, 86 and 88 respectively. The results for trial stores 77 and 88 during the trial period show a significant difference in at least two of the three trial months but this is not the case for trial store 86. We can check with the client if the implementation of the trial was different in trial store 86 but overall, the trial shows a significant increase in sales. Now that we have finished our analysis, we can prepare our presentation to the Category Manager.

[ ]:

```
```